Experiment 9

Aim: To implement Named Entity Recognition for given real world application.

Code:

NAME ENTITY RECOGNITION

A. Implementing models

1. Using Hugging Face Transformers (BERT, RoBERTa)

```
!pip install transformers datasets torch
```

```
Requirement already satisfied: transformers in
/usr/local/lib/python3.12/dist-packages (4.56.1) Requirement
already satisfied: datasets in
/usr/local/lib/python3.12/dist-packages (4.0.0) Requirement
already satisfied: torch in
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Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from transformers) (3.19.1)
Requirement already satisfied: huggingface-hub<1.0,>=0.34.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.34.4) Requirement
already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2) Requirement
already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (25.0) Requirement
already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers) (6.0.2) Requirement
already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2024.11.6)
Requirement already satisfied: requests in
/usr/local/lib/python3.12/dist-packages (from transformers) (2.32.4)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
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/usr/local/lib/python3.12/dist-packages (from transformers) (0.6.2) Requirement
already satisfied: tqdm>=4.27 in
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Requirement already satisfied: pyarrow>=15.0.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0) Requirement
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/usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16) Requirement
already satisfied: fsspec<=2025.3.0,>=2023.1.0 in
/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2025.3.0)
```

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Requirement already satisfied: typing-extensions>=4.10.0 in
/usr/local/lib/python3.12/dist-packages (from torch) (4.15.0) Requirement
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/usr/local/lib/python3.12/dist-packages (from torch) (3.5)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.1.6) Requirement
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/usr/local/lib/python3.12/dist-packages (from torch) (12.6.77) Requirement
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/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (3.12.15)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<1.0,>=0.34.0->transformers) (1.1.9)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers) (3.4.3)
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```
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests->transformers) (2025.8.3)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2.6.1) Requirement
already satisfied: aiosignal>=1.4.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.4.0) Requirement
already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (25.3.0) Requirement
already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.7.0) Requirement
already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (6.6.4) Requirement
already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (0.3.2) Requirement
already satisfied: yarl<2.0,>=1.17.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.20.1) Requirement
already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2-
>pandas->datasets) (1.17.0)
```

```
from transformers import pipeline
```

```python
# Load a pre-trained NER pipeline (BERT-based by default) ner_pipeline =
pipeline("ner", model="dslim/bert-base-NER",
grouped_entities=True)

# Example text (can replace with biomedical/financial text)

text ="Elon Musk announced a partnership between Tesla and Pfizer in California."

# Run NER
entities = ner_pipeline(text)
print(entities)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings
tab (https://huggingface.co/settings/tokens), set it as secret in your
Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks. Please
note that authentication is recommended but still optional to access public
models or datasets.
  warnings.warn(
Some weights of the model checkpoint at dslim/bert-base-NER were not used
when initializing BertForTokenClassification: ['bert.pooler.dense.bias',
'bert.pooler.dense.weight']
- This IS expected if you are initializing BertForTokenClassification from
the checkpoint of a model trained on another task or with another architecture
(e.g. initializing a BertForSequenceClassification model from a
BertForPreTraining model).
- This IS NOT expected if you are initializing
BertForTokenClassification from the checkpoint of a model that you
expect to be exactly identical (initializing a
BertForSequenceClassification model from a
BertForSequenceClassification model).
Device set to use cpu
/usr/local/lib/python3.12/dist-packages/transformers/pipelines/token_c
lassification.py:186: UserWarning: `grouped_entities` is deprecated and
will be removed in version v5.0.0, defaulted to
`aggregation_strategy="AggregationStrategy.SIMPLE"` instead.
  warnings.warn(

[{'entity_group': 'ORG', 'score': np.float32(0.9983472), 'word': 'Elon
Musk', 'start': 0, 'end': 9}, {'entity_group': 'ORG', 'score':
np.float32(0.9960098), 'word': 'Tesla', 'start': 42, 'end': 47},
{'entity_group': 'ORG', 'score': np.float32(0.9550559), 'word':
'Pfizer', 'start': 52, 'end': 58}, {'entity_group': 'LOC', 'score':
np.float32(0.9996848), 'word': 'California', 'start': 62, 'end': 72}]
```

1. Using RoBERTa for NER

```
from transformers import AutoTokenizer,
AutoModelForTokenClassification from
transformers import pipeline

# Load RoBERTa fine-tuned model for NER
model_name = "Jean-Baptiste/roberta-large-ner-english"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForTokenClassification.from_pretrained(model_name)

roberta_ner = pipeline("ner", model=model, tokenizer=tokenizer,
grouped_entities=True)

text = "Pfizer announced a new vaccine in collaboration with BioNTech in
Germany."

entities = roberta_ner(text)
print(entities)
```


{"model_id":"f22027ddfa114e1fabab2af71deb6205","version_major":2,"vers
ion_minor":0}

{"model_id":"22eba2170f1946eebca81d2f996a55c0","version_major":2,"vers
ion_minor":0}

{"model_id":"c2637b21af214ec38eeb6e8ea01523fa","version_major":2,"vers
ion_minor":0}

{"model_id":"facc7dfab5614a90acc1fbfa30e43b72","version_major":2,"vers
ion_minor":0}

{"model_id":"bbdcc102848549c29ff5f824e122d723","version_major":2,"vers
ion_minor":0}

{"model_id":"cbb723c97e9f4a429c4122fef84c3026","version_major":2,"vers
ion_minor":0}


```
Device set to use cpu

[{'entity_group': 'ORG', 'score': np.float32(0.9986677), 'word': '
Pfizer', 'start': 0, 'end': 6}, {'entity_group': 'ORG', 'score':
np.float32(0.9979841), 'word': ' BioNTech', 'start': 53, 'end': 61},
{'entity_group': 'LOC', 'score': np.float32(0.9999689), 'word': ' Germany',
'start': 65, 'end': 72}]
```

1. Using spaCy

```
!pip install -U spacy[transformers]
!python -m spacy download en_core_web_trf
```

```python
import spacy

# Load spaCy transformer-based NER model
nlp = spacy.load("en_core_web_trf")

text = "Elon Musk announced a partnership between Tesla and Pfizer on Monday in California."

doc = nlp(text)

for ent in doc.ents: print(ent.text,
      ent.label_)
```

```
Requirement already satisfied: spacy[transformers] in
/usr/local/lib/python3.12/dist-packages (3.8.7)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(1.0.13)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(2.0.11)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(3.0.10)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(8.3.6)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(2.5.1)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.1.0 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(0.4.1)
Requirement already satisfied: typer<1.0.0,>=0.3.0 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(0.17.4)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in
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```
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(4.67.1)
Requirement already satisfied: numpy>=1.19.0 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(2.32.4)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(2.11.7)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(3.1.6)
Requirement already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(75.2.0)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers]) (25.0)
Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(3.5.0)
Requirement already satisfied: spacy_transformers<1.4.0,>=1.1.2 in
/usr/local/lib/python3.12/dist-packages (from spacy[transformers])
(1.3.9)
Requirement already satisfied: language-data>=1.2 in
/usr/local/lib/python3.12/dist-packages (from langcodes<4.0.0,>=3.2.0-
>spacy[transformers]) (1.3.0)
Requirement already satisfied: annotated-types>=0.6.0 in
/usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy[transformers]) (0.7.0)
Requirement already satisfied: pydantic-core==2.33.2 in
/usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy[transformers]) (2.33.2)
Requirement already satisfied: typing-extensions>=4.12.2 in
/usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy[transformers]) (4.15.0)
Requirement already satisfied: typing-inspection>=0.4.0 in
/usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!
=1.8.1,<3.0.0,>=1.7.4->spacy[transformers]) (0.4.1)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy[transformers]) (3.4.3)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy[transformers]) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0-
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```
>spacy[transformers]) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0-
>spacy[transformers]) (2025.8.3)
Requirement already satisfied: transformers<4.50.0,>=3.4.0 in
/usr/local/lib/python3.12/dist-packages (from
spacy_transformers<1.4.0,>=1.1.2->spacy[transformers]) (4.49.0)
Requirement already satisfied: torch>=1.8.0 in
/usr/local/lib/python3.12/dist-packages (from
spacy_transformers<1.4.0,>=1.1.2->spacy[transformers]) (2.8.0+cu126)
Requirement already satisfied: spacy-alignments<1.0.0,>=0.7.2 in
/usr/local/lib/python3.12/dist-packages (from
spacy_transformers<1.4.0,>=1.1.2->spacy[transformers]) (0.9.2)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4-
>spacy[transformers]) (1.3.0)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in
/usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4-
>spacy[transformers]) (0.1.5)
Requirement already satisfied: click>=8.0.0 in
/usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0-
>spacy[transformers]) (8.2.1)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0-
>spacy[transformers]) (1.5.4)
Requirement already satisfied: rich>=10.11.0 in
/usr/local/lib/python3.12/dist-packages (from typer<1.0.0,>=0.3.0-
>spacy[transformers]) (13.9.4)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in
/usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.1.0-
>spacy[transformers]) (0.22.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in
/usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.1.0-
>spacy[transformers]) (7.3.1)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2-
>spacy[transformers]) (3.0.2)
Requirement already satisfied: marisa-trie>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from language-data>=1.2-
>langcodes<4.0.0,>=3.2.0->spacy[transformers]) (1.3.1)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=10.11.0-
>typer<1.0.0,>=0.3.0->spacy[transformers]) (4.0.0) Requirement
already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=10.11.0-
>typer<1.0.0,>=0.3.0->spacy[transformers]) (2.19.2)
Requirement already satisfied: wrapt in
/usr/local/lib/python3.12/dist-packages (from smart-
open<8.0.0,>=5.2.1->weasel<0.5.0,>=0.1.0->spacy[transformers])
```

(1.17.3)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (3.19.1) Requirement
already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (1.13.3) Requirement
already satisfied: networkx in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (3.5) Requirement
already satisfied: fsspec in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (2025.3.0)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy_transformers<1.4.0,>=1.1.2->spacy[transformers]) (12.6.77) Requirement
already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (12.6.77) Requirement
already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy_transformers<1.4.0,>=1.1.2->spacy[transformers]) (12.6.80) Requirement
already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (9.10.2.21) Requirement
already satisfied: nvidia-cublas-cu12==12.6.4.1 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy_transformers<1.4.0,>=1.1.2->spacy[transformers]) (12.6.4.1) Requirement
already satisfied: nvidia-cufft-cu12==11.3.0.4 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (11.3.0.4) Requirement
already satisfied: nvidia-curand-cu12==10.3.7.77 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (10.3.7.77) Requirement
already satisfied: nvidia-cusolver-cu12==11.7.1.2 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (11.7.1.2) Requirement
already satisfied: nvidia-cusparse-cu12==12.5.4.2 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (12.5.4.2) Requirement
already satisfied: nvidia-cusparselt-cu12==0.7.1 in
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>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (0.7.1) Requirement
already satisfied: nvidia-nccl-cu12==2.27.3 in
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already satisfied: nvidia-nvtx-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy_transformers<1.4.0,>=1.1.2->spacy[transformers]) (12.6.77)

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Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in
/usr/local/lib/python3.12/dist-packages (from torch>=1.8.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (12.6.85) Requirement
already satisfied: nvidia-cufile-cu12==1.11.1.6 in
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already satisfied: huggingface-hub<1.0,>=0.26.0 in
/usr/local/lib/python3.12/dist-packages (from
transformers<4.50.0,>=3.4.0->spacy_transformers<1.4.0,>=1.1.2-
>spacy[transformers]) (0.34.4)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from
transformers<4.50.0,>=3.4.0->spacy_transformers<1.4.0,>=1.1.2-
>spacy[transformers]) (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from
transformers<4.50.0,>=3.4.0->spacy_transformers<1.4.0,>=1.1.2-
>spacy[transformers]) (2024.11.6)
Requirement already satisfied: tokenizers<0.22,>=0.21 in
/usr/local/lib/python3.12/dist-packages (from
transformers<4.50.0,>=3.4.0->spacy_transformers<1.4.0,>=1.1.2-
>spacy[transformers]) (0.21.4)
Requirement already satisfied: safetensors>=0.4.1 in
/usr/local/lib/python3.12/dist-packages (from
transformers<4.50.0,>=3.4.0->spacy_transformers<1.4.0,>=1.1.2-
>spacy[transformers]) (0.6.2)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<1.0,>=0.26.0->transformers<4.50.0,>=3.4.0-
>spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (1.1.9) Requirement
already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=10.11.0->typer<1.0.0,>=0.3.0->spacy[transformers]) (0.1.2) Requirement
already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3-
>torch>=1.8.0->spacy transformers<1.4.0,>=1.1.2->spacy[transformers]) (1.3.0)
Collecting en-core-web-trf==3.8.0
  Using cached
https://github.com/explosion/spacy-models/releases/download/en_core_we b_trf-
3.8.0/en_core_web_trf-3.8.0-py3-none-any.whl (457.4 MB) Requirement
already satisfied: spacy-curated-transformers<1.0.0,>=0.2.2 in
/usr/local/lib/python3.12/dist-packages (from en-core-web-trf==3.8.0)
(0.3.1)
Requirement already satisfied: curated-transformers<0.2.0,>=0.1.0 in
/usr/local/lib/python3.12/dist-packages (from spacy-curated-
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transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (0.1.1) Requirement already satisfied: curated-tokenizers<0.1.0,>=0.0.9 in /usr/local/lib/python3.12/dist-packages (from spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (0.0.9) Requirement already satisfied: torch>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (2.8.0+cu126) Requirement already satisfied: regex>=2022 in /usr/local/lib/python3.12/dist-packages (from curated-tokenizers<0.1.0,>=0.0.9->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (2024.11.6) Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (3.19.1) Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (4.15.0) Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (75.2.0) Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (1.13.3) Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (3.5) Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (3.1.6) Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (2025.3.0) Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (12.6.77) Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (12.6.77) Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (12.6.80) Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (9.10.2.21) Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (12.6.4.1) Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in

/usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (11.3.0.4) Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (10.3.7.77) Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (11.7.1.2) Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (12.5.4.2) Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (0.7.1) Requirement already satisfied: nvidia-nccl-cu12==2.27.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (2.27.3) Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (12.6.77) Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (12.6.85) Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (1.11.1.6) Requirement already satisfied: triton==3.4.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (3.4.0) Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.12.0->spacy-curated-transformers<1.0.0,>=0.2.2->en-core-web-trf==3.8.0) (3.0.2)
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_trf')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option. Elon Musk PERSON
Tesla ORG
Pfizer ORG

```
Monday DATE
California GPE
```

B. Using Domain-Specific Datasets

```
from datasets import load_dataset

# Load dataset
dataset = load_dataset("financial_phrasebank", "sentences_50agree")

# Print 10 tuples (sentence + label)
for i in range(10):
    print((dataset["train"][i]["sentence"], dataset["train"][i]
["label"]))
```

```
('According to Gran , the company has no plans to move all production to
Russia , although that is where the company is growing .', 1) ('Technopolis
plans to develop in stages an area of no less than 100,000 square meters in
order to host companies working in computer technologies and
telecommunications , the statement said .', 1) ('The international
electronic industry company Elcoteq has laid off tens of employees from its
Tallinn facility ; contrary to earlier layoffs the company contracted the
ranks of its office workers , the daily Postimees reported .', 0)
('With the new production plant the company would increase its capacity to
meet the expected increase in demand and would improve the use of raw
materials and therefore increase the production profitability .', 2)
("According to the company 's updated strategy for the years 2009-2012
, Basware targets a long-term net sales growth in the range of 20 % - 40 %
with an operating profit margin of 10 % -20 % of net sales .", 2)
("FINANCING OF ASPOCOMP 'S GROWTH Aspocomp is aggressively pursuing its
growth strategy by increasingly focusing on technologically more demanding
HDI printed circuit boards PCBs .", 2)
("For the last quarter of 2010 , Componenta 's net sales doubled to EUR131m
from EUR76m for the same period a year earlier , while it moved to a zero
pre-tax profit from a pre-tax loss of EUR7m .", 2) ('In the third quarter
of 2010 , net sales increased by 5.2 % to EUR 205.5 mn , and operating
profit by 34.9 % to EUR 23.5 mn .', 2) ('Operating profit rose to EUR 13.1
mn from EUR 8.7 mn in the corresponding period in 2007 representing 7.7 %
of net sales .', 2) ('Operating profit totalled EUR 21.1 mn , up from EUR
18.6 mn in 2007 , representing 9.7 % of net sales .', 2)
```

```
!pip install datasets transformers torch
```

```
Requirement already satisfied: datasets in
/usr/local/lib/python3.12/dist-packages (4.0.0) Requirement
already satisfied: transformers in
```

```
/usr/local/lib/python3.12/dist-packages (4.56.1) Requirement
already satisfied: torch in
/usr/local/lib/python3.12/dist-packages (2.8.0+cu126)
Requirement already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from datasets) (3.19.1) Requirement
already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2) Requirement
already satisfied: pyarrow>=15.0.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0) Requirement
already satisfied: dill<0.3.9,>=0.3.0 in
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already satisfied: pandas in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2) Requirement
already satisfied: requests>=2.32.2 in
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already satisfied: tqdm>=4.66.3 in
/usr/local/lib/python3.12/dist-packages (from datasets) (4.67.1) Requirement
already satisfied: xxhash in
/usr/local/lib/python3.12/dist-packages (from datasets) (3.5.0) Requirement
already satisfied: multiprocess<0.70.17 in
/usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16) Requirement
already satisfied: fsspec<=2025.3.0,>=2023.1.0 in
/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2025.3.0)
Requirement already satisfied: huggingface-hub>=0.24.0 in
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already satisfied: pyyaml>=5.1 in
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/usr/local/lib/python3.12/dist-packages (from transformers)
(2024.11.6)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
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/usr/local/lib/python3.12/dist-packages (from transformers) (0.6.2) Requirement
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/usr/local/lib/python3.12/dist-packages (from torch) (4.15.0) Requirement
already satisfied: setuptools in
/usr/local/lib/python3.12/dist-packages (from torch) (75.2.0) Requirement
already satisfied: sympy>=1.13.3 in
/usr/local/lib/python3.12/dist-packages (from torch) (1.13.3) Requirement
already satisfied: networkx in
/usr/local/lib/python3.12/dist-packages (from torch) (3.5)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.1.6) Requirement
already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.6.77)
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Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.6.77) Requirement
already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in
/usr/local/lib/python3.12/dist-packages (from torch) (12.6.80) Requirement
already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch) (9.10.2.21) Requirement
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/usr/local/lib/python3.12/dist-packages (from torch) (12.6.4.1) Requirement
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/usr/local/lib/python3.12/dist-packages (from torch) (2.27.3) Requirement
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/usr/local/lib/python3.12/dist-packages (from torch) (12.6.77) Requirement
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/usr/local/lib/python3.12/dist-packages (from torch) (1.11.1.6) Requirement
already satisfied: triton==3.4.0 in
/usr/local/lib/python3.12/dist-packages (from torch) (3.4.0) Requirement
already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in
/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (3.12.15)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in
/usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0-
>datasets) (1.1.9)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (3.4.3)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (2025.8.3)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
```

```
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2.6.1) Requirement
already satisfied: aiosignal>=1.4.0 in
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=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.4.0) Requirement
already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
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already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
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already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (6.6.4) Requirement
already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (0.3.2) Requirement
already satisfied: yarl<2.0,>=1.17.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.20.1) Requirement
already satisfied: six>=1.5 in
/usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2-
>pandas->datasets) (1.17.0)

from datasets import load_dataset from
transformers import pipeline

# Load dataset
dataset = load_dataset("yixuantt/FinEntity", split="train")

# Inspect first sample
print(dataset[0])

# Loop over first few samples
for i in range(3): entry
    = dataset[i]
    text = entry["content"] ann
    = entry["annotations"]
```

```
    print(f"\n--- Sample {i} ---")
    print("Content:", text)
    print("Annotations:", ann)
```

```
{'content': 'Johnson & Johnson <JNJ.N> shares gained 0.20% after posting
results that beat expectations but cut its full-year outlook, citing a
stronger dollar. [nL4N2Z028U]', 'annotations': [{'end': 17, 'label':
'Positive', 'start': 0, 'tag': 'Positive', 'value': 'Johnson & Johnson'}]}

--- Sample 0 ---
Content: Johnson & Johnson <JNJ.N> shares gained 0.20% after posting results
that beat expectations but cut its full-year outlook, citing a stronger dollar.
[nL4N2Z028U]
Annotations: [{'end': 17, 'label': 'Positive', 'start': 0, 'tag': 'Positive',
'value': 'Johnson & Johnson'}]

--- Sample 1 ---
Content: On the positive side, Siemens is rallying 6% after a boom in quarterly
orders and packaging maker Huhtamaki is also up by 6% after profit beat
expectations.
Annotations: [{'end': 107, 'label': 'Positive', 'start': 98, 'tag':
'Positive', 'value': 'Huhtamaki'}, {'end': 29, 'label': 'Positive',
'start': 22, 'tag': 'Positive', 'value': 'Siemens'}]

--- Sample 2 ---
Content: Brent crude <LCOc1> rose 1.4% to $100.69 per barrel and U.S. crude
<CLc1> was also up by 1.4% to $100.61 a barrel.
Annotations: [{'end': 11, 'label': 'Positive', 'start': 0, 'tag':
'Positive', 'value': 'Brent crude'}, {'end': 66, 'label': 'Positive',
'start': 56, 'tag': 'Positive', 'value': 'U.S. crude'}]
```

```
ner = pipeline("ner", model="dslim/bert-base-NER",
grouped_entities=True)

# Take a sample
entry = dataset[1]
text = entry["content"] gold_ann
= entry["annotations"]

print("\nText:", text) print("\n
Predicted Entities:")
print(ner(text))
```

```
Some weights of the model checkpoint at dslim/bert-base-NER were not used
when initializing BertForTokenClassification: ['bert.pooler.dense.bias',
'bert.pooler.dense.weight']
- This IS expected if you are initializing BertForTokenClassification
```

```
from the checkpoint of a model trained on another task or with another
architecture (e.g. initializing a BertForSequenceClassification model
from a BertForPreTraining model).
- This IS NOT expected if you are initializing
BertForTokenClassification from the checkpoint of a model that you
expect to be exactly identical (initializing a
BertForSequenceClassification model from a
BertForSequenceClassification model).
Device set to use cpu


Text: On the positive side, Siemens is rallying 6% after a boom in
quarterly orders and packaging maker Huhtamaki is also up by 6% after
profit beat expectations.

 Predicted Entities:
[{'entity_group': 'ORG', 'score': np.float32(0.9987759), 'word':
'Siemens', 'start': 22, 'end': 29}, {'entity_group': 'ORG', 'score':
np.float32(0.98677105), 'word': 'Huhtamaki', 'start': 98, 'end': 107}]
```